Predicting Risk in Shelter Animals

Data - Louisville Metro KY - Animal Servise Intake

and Outcome

Advanced Topics in Machine Learning
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The Problem

- **Context:** Animal shelters face chronic overpopulation and limited operational resources.
- **Critical Point:** The moment of "Intake" can determine an animal's future.
- **The Problem:** High-risk animals are often identified too late for effective intervention.
- **Outcome of Interest:** Euthanasia, Death, or Disposal (Categorized as "At-Risk").

Project Goals

Primary Objective

Early binary classification of "At-Risk" vs "Safe" cases at the exact moment of shelter entry.

Uncover Profiles

Discover risk groups within the shelter population using unsupervised clustering.

Architecture Comparison

Evaluate and optimize multiple models (Logistic, Tree-based), prioritizing Recall over Precision.

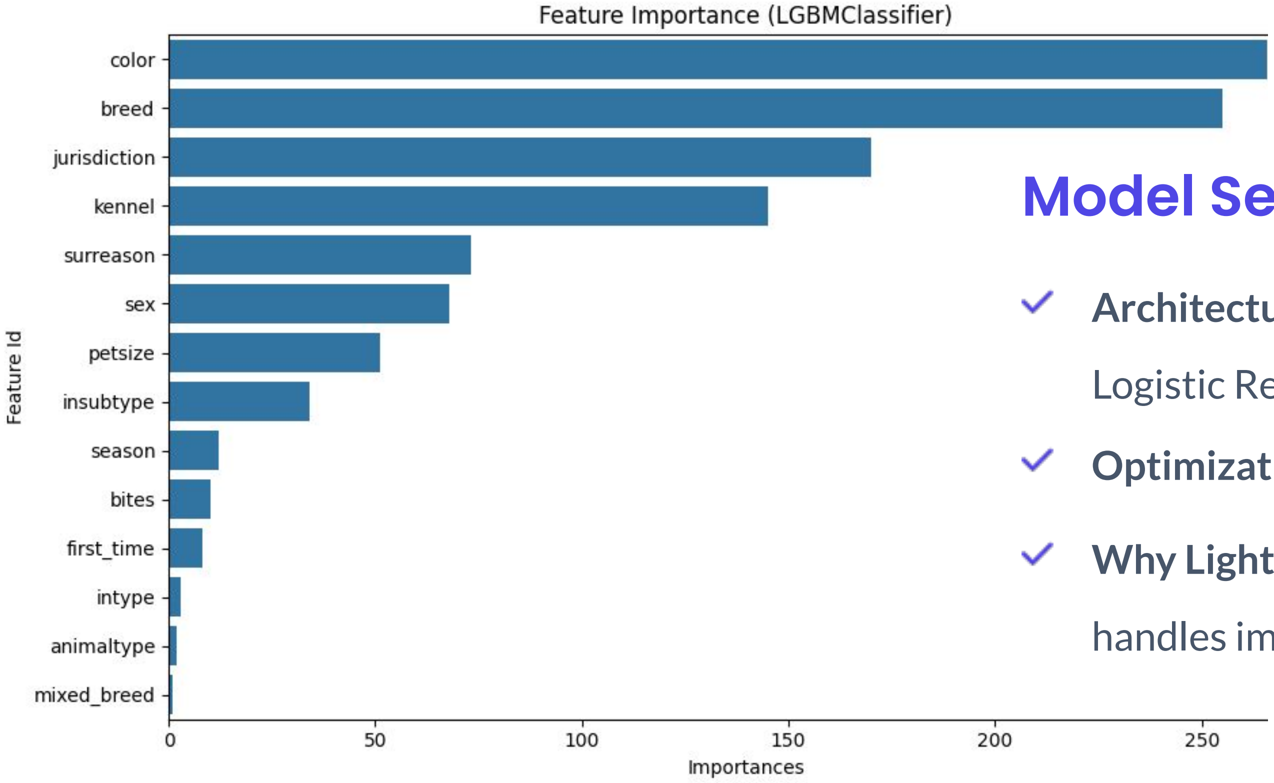
Dataset & Exploratory Analysis

Insights from Louisville Metro KY Animal Services

Data Engineering

- **Feature Parsing:** Cleaned high-cardinality breed and color data.
- **Recidivism:** Indicators for returning animals based on ID history.
- **Temporal Signals:** Seasonal features capturing influx patterns.
- **Biting History:** Critical binary flags extracted from administrative logs.

Predictive Methodology



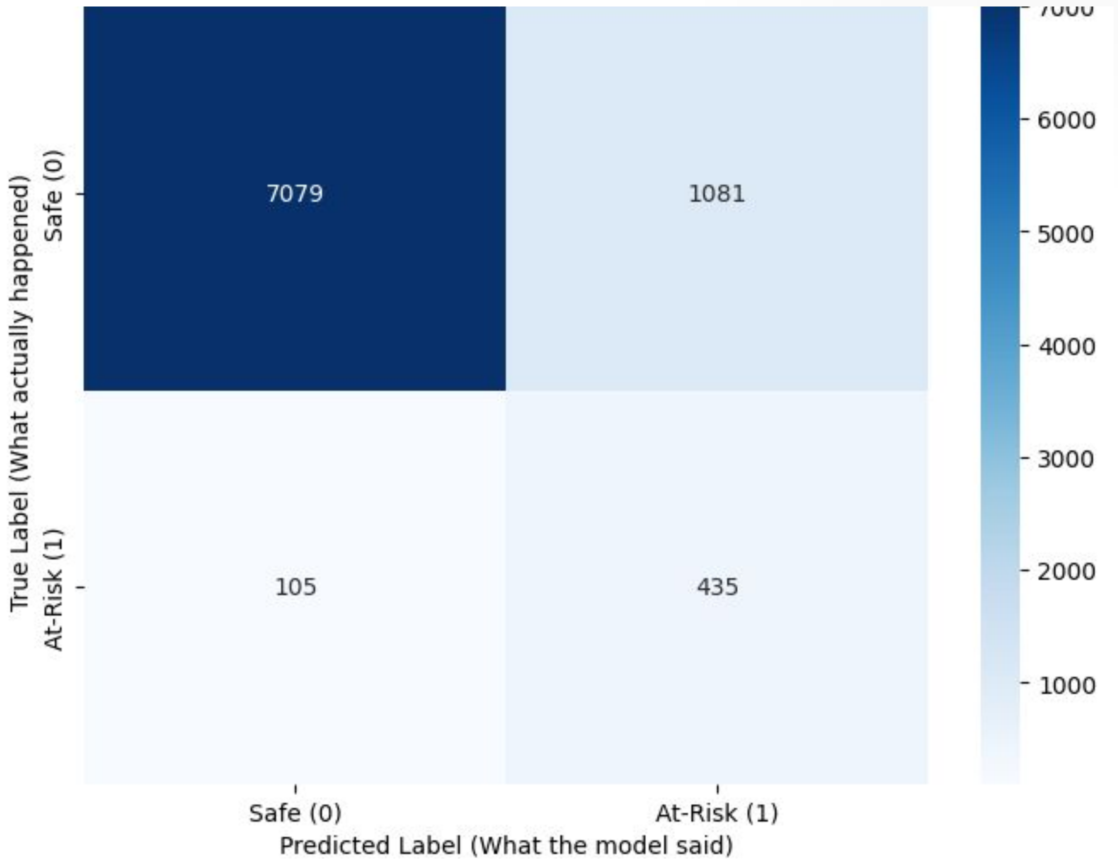
Model Selection: LightGBM

- **Architectures Evaluated:** LightGBM, CatBoost, Random Forest, MLP and Logistic Regression.
- **Optimization:** RandomizedSearchCV for hyperparameter tuning.
- **Why LightGBM?** Exceptionally efficient with categorical features and handles imbalanced data through weighted loss functions.

Performance Metrics

81.56%
Optimized Recall

Ensuring 8/10 at-risk cases are caught



Final Results Table

	Model	PR AUC	Recall	Precision	F1-Score
0	LightGBM	0.522190	0.805556	0.286939	0.423152
1	CatBoost	0.512842	0.770370	0.305209	0.437204
2	Random Forest	0.446228	0.744444	0.275342	0.402000
3	MLP	0.429660	0.479630	0.405956	0.439728
4	Logistic Regression	0.348928	0.616667	0.260156	0.365934

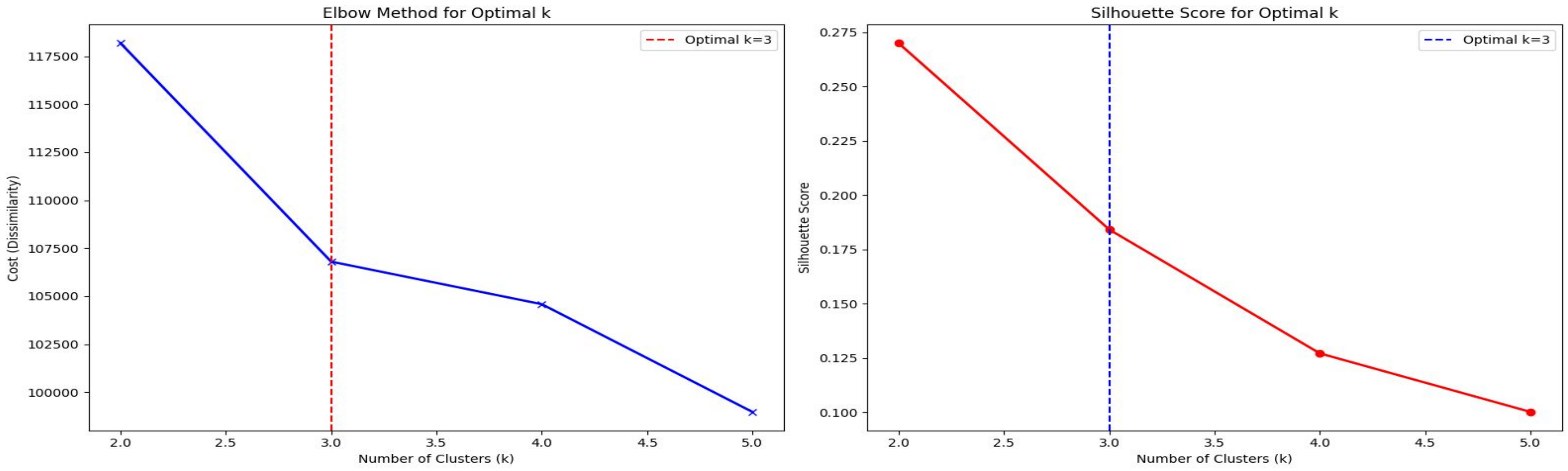
Note: Recall was prioritized to minimize the ethical cost of failing to identify a high-risk animal at intake.

Unsupervised Risk Profiles

K-Modes Clustering

Standard K-Means fails on categorical shelter data. K-Modes provides robust similarity matching.

- **Elbow Method:** Used to define optimal risk segments (k=10).
- **Outcome:** Clear stratification of high and low mortality clusters.



Comparison & Discussion

The Ethical Cost of Error

A "False Negative" in this domain is irreversible. Missing a high-risk animal leads to missed behavioral and medical triaging that could have saved its life.

Mission Alignment: The model functions as a safety net, ensuring the most vulnerable shelter residents receive immediate prioritization.

Conclusions & Future Work

- **Proactive Management:** Shifts shelter operations from reactive response to data-driven early intervention.
- **NLP Integration:** Future work involves analyzing text-based medical and behavior notes for deeper insights.
- **Real-time API:** Planning for deployment as a intake-desk decision tool for staff.

Questions?

Thank you for your attention.

Image Sources

https://www.fortworthtexas.gov/files/assets/public/v/1/code-compliance/animals/images/surrender.jpg?dimension=pageimagefullwidth&w=1140

Source: www.fortworthtexas.gov

https://ucarecdn.com/a4851ecf-bde3-4cea-a6cb-1f72294ab5a7/-/resize/960x/-/format/auto/

Source: www.houstonhumane.org

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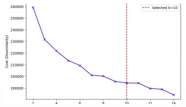

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